Name: _____ **MTH234 Linear Algebra**
ID: _____ **1/2025 (Regular/International Section)**
22nd October 2025
3 hrs. (09:00 - 12:00 AM.)

MODULE 2 MOCK EXAM (Total 50 points)

Instruction: Students must provide all details of your work. You may use any class materials. Calculators and other materials are not allowed.

(1) (6 points) Let $V = P_2$ be the vector space of all polynomials of degree at most 2. Let W be the subset of V defined by

$$W = \{p(x) = ax^2 + bx + c \in P_2 \mid 3a - 2b - c = 0\}$$

Show that W is a subspace of P_2.

(2) (8 points) Let $V = P_2$ be the vector space of all polynomials of degree at most 2. Determine whether the polynomial $p(x) = 1 - 4x + 6x^2$ is in the span of the set $S = \{p_1(x), p_2(x), p_3(x)\}$, where

$$p_1(x) = 1 + x - x^2, \quad p_2(x) = 2 + x, \quad p_3(x) = -1 + x - 3x^2$$

(3) (8 points) Let the matrix A be given by

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 4 & 1 & 1 \\ 3 & 6 & 1 & 0 \end{bmatrix}$$

By reducing the matrix to its row echelon form, determine the following:

(a) A basis for the Row Space of A, and state its dimension.

(b) A basis for the Column Space of A, and state its dimension.

(c) A basis for the Null Space of A, and state its dimension (the nullity).

(4) (10 points) Let B and B' be two bases for the vector space P_2:

$$B = \{p_1(x), p_2(x), p_3(x)\} = \{1, 1+x, 1+x+x^2\}$$
$$B' = \{q_1(x), q_2(x), q_3(x)\} = \{1-x, x-x^2, 2x^2\}$$

(a) Find the transition matrix ${}_{B'}P_B$ from the basis B to the basis B'.

(b) For the polynomial $p(x) = 3 + x - 2x^2$, find its coordinate vector $[p(x)]_B$, which is coordinate vector of $p(x)$ with respect to the basis B.

(c) Find the coordinate vector $[p(x)]_{B'}$, which is coordinate vector of $p(x)$ with respect to the basis B'.

(5) (8 points) Let the matrix A be given by

$$A = \begin{bmatrix} 1 & -2 & 1 & 3 & 0 \\ 2 & -4 & 3 & 7 & -1 \\ -1 & 2 & 1 & -1 & -2 \\ 3 & -6 & 2 & 8 & 1 \end{bmatrix}$$

(a) Find the rank of A.

(b) Find the nullity of A.

(c) Explicitly verify that the Rank-Nullity Theorem holds for this matrix by finding the rank and nullity from their definitions.

(6) (10 points) Let W be a subspace of $\mathbb{R}^4$ with the basis $S = \{w_1, w_2, w_3\}$, where

$$w_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad w_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \quad w_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

(a) Use the Gram-Schmidt process to find an orthonormal basis for W.

(b) Using the basis found in part (a), find the orthogonal projection of the vector $v = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ onto the subspace W.